

Customer Information

Freeheart Properties

+233200000000

freeheart.abokobi@example.com

Freeheart, Abokobi, Greater Accra

Abokobi, Ghana

Quote #: QT-FREEHEART-831A93

Date: May 15, 2026

Valid for: 30 days

Hybrid Solar Package

Hybrid solar for ~65 kWh/day at Abokobi (Accra sun data). 39× 570W panels, 20.0 kW hybrid inverter, 20.0 kWh storage (8h essential backup).

Item	Unit price (GHS)	Qty	Amount (GHS)
Longi 570W Panel	1,500.00	39.0	58,500.00
PV mid / end clamp (each) (3.5× panels)	30.00	136.0	4,080.00
MC4 connector pair (~1.2 pairs per panel)	25.00	47.0	1,175.00
Energy Precision 10kW 20.0kW Inverter	11,000.00	1.0	11,000.00
RR 10MM Copper Cable (100m coil) (AC feeder coils vs inverter kW)	2,500.00	2.0	5,000.00
125A AC breaker (inverter 10.0 kW > 8 kW)	300.00	1.0	300.00
AC isolator switch 40A (AC isolator per inverter)	280.00	1.0	280.00
100A changeover (1 per inverter / genset path)	1,500.00	1.0	1,500.00
Energy Precisions 16.0kWh Battery	26,000.00	2.0	52,000.00
Battery cable 50mm copper (per meter) (~0.9 m per kWh (large bank))	200.00	29.0	5,800.00
Battery cable 35mm copper (per meter) (supplementary 35 mm runs)	80.00	16.0	1,280.00
250A MCCB battery breaker (battery bank protection)	650.00	4.0	2,600.00
Mounting structure (rails) (proforma rail sets)	350.00	25.0	8,750.00
Transportation of logistics	3,000.00	1.0	3,000.00
Installation & commissioning (standard) (tier for 22.2 kW)	6,000.00	1.0	6,000.00
	2,500.00	2.0	5,000.00

Item	Unit price (GHS)	Qty	Amount (GHS)
Single core 6mm PV cable (100m coil) (PV homerun (4 DC strings))			
AC cable 6 mm² (100m coil) (AC branch / DB wiring)	1,800.00	1.0	1,800.00
63A DC MCB 2P (DC string groups (4 strings))	200.00	4.0	800.00
DC isolator switch 32A (DC isolator per string group)	350.00	4.0	1,400.00
100A AC breaker (main AC feed)	250.00	1.0	250.00
6-way breaker box	180.00	1.0	180.00
13-way breaker box	350.00	3.0	1,050.00
SPD / lightning arrester (500V class) (standard hybrid)	300.00	1.0	300.00
Earthing kit (rod, clamp, 16mm earth cable) (earthing per array section)	450.00	4.0	1,800.00
PVC trunking 25mm (per metre) (trunking metres)	35.00	59.0	2,065.00
DC/AC labeling & warning kit (labels & warnings)	120.00	1.0	120.00
Miscellaneous (larger kit — Option 2 reference) (remaining consumables (glands, lugs, ties))	2,000.00	1.0	2,000.00

Sub total: 178,030.00

Tax (12.5%): 22,253.75

Discount (0%): 0.00

Option total (GHS): 200,283.75

SYSTEM SPECIFICATIONS & ENGINEERING DETAILS

DAILY ENERGY
REQUIREMENT

65.12 kWh/
day

EFFECTIVE DAILY
ENERGY

90.45 kWh/
day

After system losses

MONTHLY
GENERATION

2713.5 kWh

PEAK SUN HOURS

5.0 hrs/day

Location: N/A

PV ARRAY
CAPACITY

22.23 kW

39 × 570W panels

PANEL BRAND

Longi

570W model

INVERTER
CAPACITY

16.7 kW

Selected: 20.0kW
(for quote)

DC/AC RATIO

1.11

Optimized

BATTERY
CAPACITY

20.0 kWh

BACKUP
DURATION

8 hours

REQUIRED ROOF
AREA

121.7 m²

MOUNTING RAILS
(EST.)

20 sticks (18 ft)

≈ 105.3 m linear

SYSTEM
EFFICIENCY

72.0 %

Engineering Notes: This system is designed using industry-standard engineering factors, including system efficiency (accounting for inverter, wiring, and temperature), an optimized DC/AC ratio to prevent inverter clipping, and location-specific peak sun hours for accurate energy-generation calculations. All calculations include appropriate safety margins for reliable performance.

Mounting rail stick count is a **planning estimate** (18 ft segments, portrait grid); confirm with site layout and structural design.

Terms & Conditions

This quotation is valid for 30 days from the date of issue. All prices are subject to change without notice. Installation and project delivery terms will be discussed upon acceptance.

Payment Terms

30% deposit · 40% on delivery · 30% on commissioning

Notes

Site: Freeheart, Abokobi (Greater Accra). Buildings: 3-storey apartment (L/R wings per floor) + 3-storey glass building. Load basis: client point count — lights 78, outlets 75, fans 30. Assumptions: 12W LED, 150W diversified outlet load, 70W fans. Contact placeholder: Site Contact — Abokobi +233200000000

Thank you for your business!